The Impact of Expanding Public Health Insurance on Safety Net Program Participation: Evidence from the ACA Medicaid Expansion

by

Lucie Schmidt
Department of Economics, Smith College
Department of Economics, Williams College
National Bureau of Economic Research

Lara Shore-Sheppard
Department of Economics, Williams College
National Bureau of Economic Research

Tara Watson
Department of Economics, Williams College
National Bureau of Economic Research
Brookings Institution

CES 26-32 **May 2026**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

Papers in the series are posted on www.census.gov/library/working-papers/series/ces-wp.html. For information about the series, contact Danielle H. Sandler, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov.

Abstract

We examine spillover effects from the ACA Medicaid expansion to public programs providing cash and food assistance. We consider program participation in contiguous county pairs crossing state borders, where one state took up the Medicaid expansion and the other did not, allowing us to better control for local economic trends that could affect program participation. We find that the Medicaid expansion increased participation in food assistance and one of the cash programs, with impacts mainly due to participation conditional on eligibility, rather than from labor supply responses. Our results demonstrate the potential for spillovers across safety net programs.

* We are grateful to the Russell Sage Foundation for financial support and to Nadine Nichols at the U.S. Department of Agriculture and to Brian Stacy at the World Bank for data. We received helpful comments from Marianne Bitler, Michael Dworsky, Emily Lawler, Tom Selden, Jeffrey Smith; conference participants at the American Society of Health Economics Annual Meeting, the Association for Public Policy and Management Annual Meeting, and the Southern Economic Association Annual Meeting; and seminar participants at the Center for Health Economics and Policy Studies at San Diego State University, the Duke University Population Research Institute, George Mason University, Hunter College, the University at Albany, the University of Maryland Population Research Center, the University of Massachusetts-Amherst, the University of South Florida, the University of Wisconsin Center for Demography and Ecology, and Williams College. Louisa Abel, Mary Beth Dato, Casey DeLano, Gabriela Goodman, Huy Nguyen, and Ahna Pearson provided excellent research assistance. Michael L. Anderson offered useful guidance on standard errors. Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2192. (CBDRB-FY24-P2192-R11194, CBDRB-FY24-P2192-R11157, CBDRB-FY23-P2192-R10563)

I. Introduction

The U.S. safety net is a nexus of public programs addressing a partially overlapping set of needs and serving a partially overlapping set of beneficiaries. Understanding how these programs interact can provide lessons for facilitating access and can improve program design. For example, a change in eligibility for one program might facilitate information transmission about other programs, might reduce the administrative burden associated with applying for other programs, or might directly impact eligibility for other programs. Eligibility could also be impacted by a labor supply response to changes in another program. Our work provides a case study of program spillovers using the Affordable Care Act (ACA) Medicaid expansion, which significantly expanded public health insurance eligibility in participating states, as a natural experiment.

Understanding how participation in non-health safety net programs responds to public health insurance expansions furthers the ability to address barriers to program take-up. In addition, spillovers between programs affect the implied costs and benefits of the Medicaid expansion from both the perspective of the states as well as that of the federal government. This study therefore contributes to the policy evaluation of the ACA, and can inform future state and federal expansion decisions. Finally, there is a vast literature examining the effects of the ACA Medicaid expansion on a large number of outcomes.[1] If the Medicaid expansion spilled over onto participation in other programs, that has important implications for the interpretation of those studies and the implied mechanisms underlying the estimated effects.

The ACA expanded the availability of the Medicaid public health insurance program to individuals who were previously categorically ineligible, as well as to individuals such as parents who had been categorically eligible but at lower means-tested income limits. Given connections across programs in eligibility criteria, information flows, and implementation, this major change in Medicaid eligibility may have impacted participation in other programs. In this paper, we explore the impact of expanded Medicaid access through the ACA on participation in the two most widely used and broadly available non-health safety net programs: the Earned Income Tax Credit (EITC) and the Supplemental Nutrition Assistance Program (SNAP), as well as on

[1] For example, Soni, Wherry, and Simon (2020) review 43 studies examining the effects of the ACA Medicaid expansion on health outcomes. These studies find evidence of improvements in health status, maternal and neonatal health, chronic disease, and mortality.

participation in the considerably smaller and more restrictive Temporary Assistance to Needy Families Program (TANF).

Expansion of Medicaid eligibility has several potential impacts on participation in other safety net programs. First, changes in earnings due to the new Medicaid income limits could affect eligibility. Individuals may be induced to reduce their income below the limit to qualify for Medicaid. Alternatively, the newly increased income limit relative to previous limits for some groups could allow individuals to increase earnings while still qualifying for Medicaid. Either reductions or increases in earnings could imply changes in EITC, SNAP, and TANF program eligibility, as outlined below. However, previous literature has found little evidence of employment or earnings effects of the ACA Medicaid expansion (Duggan, Goda, and Jackson 2019).

In addition, expanding Medicaid access may facilitate participation in other programs by reducing administrative burden–that is, difficulty learning about or accessing resources that is created by program design features (Herd and Moynihan 2018; Herd et al. 2023). The Medicaid expansion could draw individuals' attention to the existence of, and their potential eligibility for, other public programs. It could also reduce the marginal cost of enrolling in additional programs. Individuals incur informational and administrative costs while enrolling in Medicaid and therefore may find it comparatively straightforward to add one or more safety net programs to their benefits package. Cross-program spillovers may be more likely when states make it easy to qualify for and enroll in such programs, or when a state has many residents who are otherwise disconnected from the safety net.

In this paper, we follow much of the state difference-in-differences literature that exploits the 2012 Supreme Court decision that made the Medicaid expansion optional for states. Unlike that literature, however, we identify the effect of the Medicaid expansion by comparing changes in participation in contiguous pairs of counties that cross state lines, where the county on one side of the border experienced the Medicaid expansion and the county on the other side did not. This approach allows us to focus narrowly on differences arising from the ACA Medicaid expansion choice, abstracting from potential heterogeneity in trends across broader geographic areas. We use county-level administrative data, which has the benefit of not being subject to under-reporting bias, as well as individual-level restricted-access American Community Survey

(ACS) data, which has the advantage of allowing analyses of heterogeneity across groups and of mechanisms.

Using a border-pair method, we begin by presenting evidence of increases in insurance coverage due to the ACA Medicaid expansion that are of magnitudes similar to those found by other authors using state difference-in-differences and state synthetic control methods (Kaestner et al. 2017, Courtemanche et al. 2017). We then turn to county-level administrative measures of EITC claiming and SNAP participation. Our estimates of the effect on EITC claiming are fairly imprecise, but are suggestive of small increases in the number of tax returns claiming the EITC. For SNAP, we find that the Medicaid expansion led to a statistically significant increase in participation.

We next use individual-level restricted-access ACS data to directly examine joint participation in Medicaid and SNAP and participation in TANF, as well as to explore heterogeneous effects and possible mechanisms. The ACS analysis confirms significant effects of the Medicaid expansion on health insurance coverage and SNAP participation overall, across demographic groups, and within the low-income population. Our results suggest that a fifth of those induced to obtain public insurance as the result of ACA expansions are also induced to participate in SNAP. We also find positive impacts on participation in TANF, a program we are unable to analyze using administrative county-level data. It does not appear that the program participation increases are mainly due to labor supply changes. We therefore conclude that the safety net participation response is primarily driven by increased information or reductions in transaction costs.

Finally, we examine how the state policy environment affects the extent of ACA expansion spillovers to SNAP participation. States and localities with more relaxed SNAP eligibility rules and states with lower SNAP take-up prior to ACA implementation see bigger increases in SNAP following the ACA Medicaid expansions. Our results suggest that access to Medicaid lowered the marginal cost of enrolling in other safety net programs and facilitated SNAP participation among the low-income population, particularly in areas with higher initial SNAP unmet need.

II. Background

A. The Medicaid Expansion and Other Safety Net Programs

The 2010 Affordable Care Act was arguably the most important piece of health care legislation in a generation. It aimed to reduce uninsurance by mandating that individuals obtain health insurance, by improving the functioning and regulation of the private market, and by significantly expanding access to public health insurance through Medicaid. In particular, it sought to expand eligibility for Medicaid to all those under 138 percent of the federal poverty level (FPL), representing a major change in most states. The change was especially pronounced for adults without dependents or disabilities who had been largely excluded from the Medicaid program. In 2012, the Supreme Court ruled that states could not be compelled to participate in the Medicaid expansion, leading to substantial variation in the extent and timing of Medicaid expansions across states. Many states expanded in January 2014, but some expanded earlier or later, and ten states have not (as of December 2023) changed their Medicaid eligibility limits.

As noted above, many individuals affected by the Medicaid expansion are potentially eligible for other key safety net programs: EITC, SNAP, and TANF. As a refundable tax credit to low-income workers, eligibility for the EITC depends on the income a family earns. The amount of the credit rises as a family's income rises (the "phase-in") until the maximum credit has been attained, at which point the amount of the credit stays constant as income rises (the "plateau"). Above a cutoff amount of income, the amount of the credit falls as income increases (the "phase-out") until the credit falls to zero. The maximum amount of the credit depends on the number of children in the family, and the earned income limit varies depending on filing status.[2] Participation in the program among eligible taxpayers is relatively high, with overall take-up rates estimated to be between 75 and 79 percent (Nichols and Rothstein 2016), although take-up rates vary according to demographics, family structure, and position on the EITC schedule (Jones 2014), and incomplete take-up of the EITC remains a concern among policymakers (National Conference of State Legislatures 2018; Internal Revenue Service 2019). Many states also supplement the EITC, generally by providing additional support for those in the Federal EITC eligibility range.

[2] In 2014, for example, the maximum credit was $496 for a family with no qualifying children, $3,305 with one qualifying child, $5,460 with two qualifying children, and $6,143 with three or more qualifying children, and the earned income limit ranged from $14,590 for a single individual with no children to $52,427 for a married couple with three or more children (Internal Revenue Service 2013).

Unlike the EITC, the structure of SNAP is that of a traditional means-tested transfer program: food benefits (awarded in the form of Electronic Benefit Transfer cards that can be used to purchase food from grocery stores) are highest among families with no earnings and are reduced at a 30 percent rate as earnings rise. To be eligible, a household must have gross monthly income below 130 percent of the poverty level and "countable" assets below $2,250 (Hoynes and Schanzenbach 2016); these rules are somewhat relaxed in states that have adopted "broad-based categorical eligibility" criteria which allow states to relax gross income and asset requirements. SNAP is one of the few safety net programs that is not limited to certain categories of low-income households, although nonworking adults up to age 49 without children or disabilities are limited to three months of benefits within a three-year period (Hoynes and Schanzenbach 2016). Similar to the EITC, overall take-up rates are relatively high for a safety net program—75 percent in 2010—but they vary across states and groups of individuals within states (Cunnyngham 2012).

TANF provides means-tested cash benefits to low-income families with dependent children. Traditionally, TANF and its predecessor, Aid to Families with Dependent Children, primarily served single parent families, although two-parent families and child-only cases now make up a larger share of the caseload (Ziliak 2016). States set income eligibility limits for TANF, and limits are generally much lower than those for SNAP or for Medicaid under the ACA expansion. The median state income eligibility limit was at 48 percent of the federal poverty level (Falk 2014), and of every 100 poor families with children in 2019, only 23 received TANF (Pavetta, Safawi, and Trisi 2021).

B. Potential Mechanisms for Safety Net Program Spillovers

Many of the possible interactions between expanded Medicaid eligibility and participation in other programs arise from interactions between eligibility and labor market decisions of workers. These decisions may be either on the extensive margin (enter or leave the labor market) or on the intensive margin (work more or fewer hours conditional on working at all). The likely impacts also differ by whether the individual is single or married and by the presence or absence of children. In addition, any spillovers from Medicaid expansion to the other safety net programs due to changes in labor supply will depend on the relationship between the Medicaid income eligibility limit and the limits for each program.

Single and married parents had a route onto Medicaid prior to the ACA, but in most states needed to have very low incomes to qualify. Some parents may have chosen to stay out of the labor force to obtain health insurance; single parents were probably more likely to be on this margin because their families were less likely to have an employer-sponsored option. With a higher Medicaid income eligibility threshold post-expansion, parents could enter the labor force without losing Medicaid coverage. Entry into employment would likely increase EITC eligibility because non-earners cannot receive the EITC. If earnings are below the generally low TANF threshold, labor market entry could potentially raise access to TANF in the presence of work requirements. SNAP eligibility is unlikely to be affected by entry into employment because parents would typically qualify with any level of income up to 130 percent of the federal poverty level.

Another group of parents would earn above the Medicaid threshold of 138 percent of the federal poverty line in the absence of ACA expansion, and could be induced to reduce their labor supply to qualify under the ACA. With a moderate reduction in earnings, these parents would also be likely to qualify for SNAP which has a similar threshold. They would be less likely to become newly eligible for TANF, because those income limits are much lower, and EITC eligibility is unlikely to be impacted because of the high income eligibility thresholds for parents.

The predictions are different for working-age adults without children, who were typically ineligible for Medicaid at any level of income prior to the ACA unless they had a disability. Therefore, there is no predicted extensive margin effect. Non-parents who were earning above 138 percent of the FPL in the absence of expanded Medicaid now have an incentive to work less to qualify for Medicaid in expansion states. A non-parent who reduces labor supply to gain eligibility for Medicaid might also now be eligible for SNAP and the EITC. Non-parents are typically ineligible for TANF.

In addition to labor market-related effects, there may be reasons related to information or transaction costs for a change in participation in any of these programs. Workers with low incomes who would not normally file a tax return may learn about EITC benefits or be nudged to file taxes when enrolling in Medicaid.[3] It is also possible that changes in labor supply induced by

[3] The threshold at which an individual or family is required to file taxes is in all cases below the expanded Medicaid eligibility cutoff.

the ACA put a low-income worker in a position on the EITC schedule where it is worth the effort to file a return, even if that worker would have been eligible for a small EITC benefit previously.

Gaining access to Medicaid may increase individuals' awareness of the possibility of eligibility for SNAP or TANF, or make the application process easier. Often, government offices or non-governmental organizations that assist with Medicaid enrollment offer information about or assistance with applications for SNAP, for example. Finkelstein and Notowidigdo (2019) show that information and assistance can increase SNAP enrollment. These effects are likely to be larger in places where there is more unmet need, which we proxy by low baseline SNAP participation rates among the eligible population.

In sum, if expanded Medicaid access changes labor supply by promoting labor market entry at a level below the new Medicaid cutoff or by reducing labor supply on the intensive margin to qualify, the most likely result would be to increase eligibility for the EITC, SNAP, and TANF. The information/transaction costs channel would also promote take-up of other programs conditional on eligibility, particularly in areas with more unmet need among low-income individuals. The magnitude of any effect of Medicaid eligibility expansion on safety net participation is ultimately an empirical question.

C. Previous Research on Program Spillovers

The interconnected nature of safety net program eligibility and receipt makes examining the impacts on other programs an important aspect of evaluating the effect of changes in any one program. However, there has been relatively little work on the impact of recent Medicaid expansions on cash and food program participation. Baicker et al. (2014) analyze the Oregon Health Insurance Experiment, in which permission to apply for Medicaid was randomly assigned among a set of low-income adults. They find no effects on TANF receipt, but they find small positive effects on the probability of SNAP receipt. Burney, Boehm, and Lopez (2021) use data on SNAP participation from the Consumer Expenditure Survey and a state difference-in-differences specification that exploits the variation in state ACA Medicaid expansion decisions. They report positive impacts of the ACA Medicaid expansion on SNAP participation. Cha and Escarce (2022) also use a state difference-in-differences specification and data from the Current Population Survey and find a 2.9 percentage point increase in SNAP among the low-income population. Barcellos, Jacobson, and Levy (2023) use a regression discontinuity design at the 138 percent FPL cutoff in expansion states, and while their focus is primarily health spending

and health insurance, they also look for effects on SNAP, energy assistance, public housing, and unemployment insurance, and find none. To our knowledge, no prior work examines how the ACA Medicaid expansions affected EITC or TANF receipt.

III. Data

A. Medicaid Income Limits and Forming County Border Pairs

Prior to the ACA, parents could receive Medicaid if their incomes were below the parental eligibility limit, and a very few states had limited eligibility for non-parents. After the ACA-related Medicaid expansions, income limits were raised for adults regardless of parental or disability status in a number of states. In our empirical work, we focus on the maximum non-categorical income limit—the limit facing able-bodied non-parents. Our primary sources for Medicaid income eligibility levels are reports published by the Kaiser Family Foundation (Cohen Ross et al. 2009, Heberlein et al. 2011, Heberlein et al. 2012, Heberlein et al. 2013, Brooks et al. 2015, Brooks et al. 2016, Kaiser Commission on Medicaid and the Uninsured 2013) and the Urban Institute's TRIM3 program rules database supplemented by information from state plan amendments available from the Centers for Medicare and Medicaid Services and state websites.

We determine which counties are contiguous using two files from the Census Bureau: a 2015 county adjacency file, which lists all adjacent counties, regardless of type of adjacency, and a county adjacency file from 1991 which gives the type of adjacency. We adjust the 2015 county-pair list to keep only counties that share a common land border or that are separated by a body of water but connected by a bridge or boat.[4]

Figure 1 illustrates the variation we use in our empirical approach: the sub-state divisions shown are counties, and contiguous border county pairs that differed in their Medicaid expansion status as of April 2014 are highlighted. In 2014, there were 488 contiguous county pairs in which one county was in a state that had adopted the Medicaid expansion and the other in a state that had not, out of a total of 1195 contiguous county pairs. There are also two sources of variation used in our analysis that are not shown in Figure 1. First, a few states had already expanded eligibility for Medicaid to some adults without disabilities prior to the ACA expansion, so the

[4] We eliminate counties that meet at a corner only and counties that are separated by a body of water and have no direct bridge or boat connection.

ACA expansion increased eligibility for Medicaid more in some states than in others.[5] Second, states varied in when, as well as in whether, they expanded Medicaid. The modal expansion date (21 states) was January 1, 2014, but some states began to expand in 2010 and other states did not expand until later in 2014 or in later years.[6] Our analysis considers expansions through 2016.[7]

B. County-Level Administrative Data

Our data on health insurance coverage at the county level come from the Census Bureau's Small Area Health Insurance Estimates (SAHIE) program, which produces estimates of the fraction with and without health insurance coverage by income group at the county level. The SAHIE estimates are model-based, incorporating information from the American Community Survey, federal tax return data, data on Supplemental Nutrition Assistance Program caseloads, Medicaid and Children's Health Insurance Program caseloads, Census population estimates, County Business Patterns, and the 2010 Census.

Recipients claim the EITC via income tax filing, and the Internal Revenue Service reports the number of returns filed in each county in each year with the Earned Income Credit as part of the Statistics of Income administrative records of individual income tax returns (Forms 1040). We use data for the 2010-2016 tax years (i.e. returns filed in 2011-2017). We adjust for county size by denominating the count of returns by estimates of the working age (18-64) population from the Census Bureau, and multiply by 100 to get the number of EITC returns per 100 working age adults.

For SNAP, we use data on the number of recipients by county from the US Department of Agriculture Food and Nutrition Service National Data Bank. These data are reported biannually, in January and July, and we use the January counts for 2011-2017 to describe participation at the end of the prior year. Some states only report SNAP participation at the state

[5] California rolled out its early Medicaid expansion on a county-by-county basis. We include this variation, which results in additional discordant pairs on the California state border prior to 2014; beginning in 2014, all states bordering California also expand, so these pairs are no longer discordant at that point.

[6] Recent literature points out potential biases generated by difference-in-differences models with staggered treatment timing (Goodman-Bacon 2021). We show that our results are robust to these issues below.

[7] Of 1195 contiguous border county pairs, 746 have discordant values for the non-categorical income limit at some point between 2010 and 2016. The number of discordant pairs by this definition ranges from 109 in 2010 to 562 in 2014.

level, so we are unable to use county-level data for the states of Connecticut, Idaho, Maine, Massachusetts, Missouri, Montana, Nebraska, New Hampshire, New York, Oregon, Rhode Island, Utah, Vermont, Washington, West Virginia, and Wyoming. We adjust for county size by denominating the count of SNAP participants with total population estimates in each county from the US Census Bureau, and multiply by 100 to get a SNAP participation rate per 100 residents. Our county level control variables include the shares of the county population that are non-Hispanic Black and Hispanic from the Census Bureau. In robustness checks, we also control for the unemployment rate, which we obtain from the Bureau of Labor Statistics' Local Area Unemployment series.

Table 1 shows variable means for both the all-county and contiguous-county samples for both the EITC and SNAP data, and in addition shows the first year (2010) and last year (2016) separately. EITC and SNAP participation are similar in the sample of all counties and in the contiguous counties sample. The counties in the border sample have somewhat fewer Hispanic residents and somewhat more non-Hispanic Black residents. Between 2010 and 2016, the number of EITC returns increased while the SNAP participation rate fell, consistent with the improving economy. In both EITC and SNAP data, the border county sample is more likely to have expanded Medicaid, and as a result the average non-categorical income limit is higher in the border county sample. To illustrate the compositional effects of using the border pair sample rather than all counties, we estimate the county-level models on the all-county sample and the contiguous county sample before incorporating the within-pair controls.

C. Individual-Level Data from the American Community Survey

Though county-level administrative data has the advantage of completeness and accuracy, it does not allow us to investigate the mechanisms underlying increased safety net participation nor to examine participation in multiple programs or heterogeneity in impacts. Thus, we conduct additional analyses using individual-level data from the American Community Survey (ACS), using restricted-access data because the public-use ACS does not identify county of residence. The ACS has questions about health insurance, SNAP participation, income, and labor supply. In addition, the ACS asks about receipt of TANF, a safety-net program without available county-level administrative data. The ACS does not ask about EITC receipt, so we impute EITC eligibility using information on marital status, income of the individual and their spouse or unmarried partner in the household, number and ages of children in the household and

the federal EITC rules in the relevant year. Because the SNAP and TANF income questions refer to the twelve months prior to the survey, we use the prior year's Medicaid income limit and focus on survey years 2011-2017.[8]

We examine all adults ages 25 to 64 in border counties, and then we split the sample into four demographic groups based on parental status and age, which are characteristics that affect the program eligibility rules faced by an individual. Parents, who faced somewhat less restrictive Medicaid eligibility criteria pre-ACA, are defined as those with a biological, step, or adoptive child under 19 in the home, or unmarried partners of those with children in the home. Among parents, we split the sample between married and unmarried, given the differing EITC rules for these groups. Among non-parents, we split the sample into those who are ages 25 to 49 and therefore potentially affected by SNAP work requirements for able-bodied adults without dependents, and those ages 50 to 64. We end up with four demographic subgroups: younger non-parents, older non-parents, married parents, and single parents.

Summary statistics are reported in Table 2. Of the four groups, single parents have the highest rates of safety net program participation, which is unsurprising both because their incomes are lower on average and because safety net programs typically have less restrictive eligibility for parents than for childless adults. However, rates of participation in SNAP among the apparently income-eligible are relatively low even for single parents, with only 65 percent of single parents with incomes below 130 percent of the poverty line reporting SNAP participation. Notably, over 70 percent of all groups report non-zero usual work hours, with the rate over 80 percent for younger non-parents and both single and married parents.

IV. Empirical Approach

Consider the following specification estimated on a sample of all counties in the continental U.S. for the period 2010-2016:

$$(1) \qquad y_{ct} = \alpha + \delta \text{NoncategoricalLimit}_{s(c)t} + X_{ct}\Gamma + \varphi_c + \tau_t + \varepsilon_{ct}$$

where y_{ct} denotes the various outcomes of interest (described in detail below) for county c in time t, where t denotes year. $NoncategoricalLimit_{s(c)t}$ denotes the Medicaid non-categorical

[8] The American Community Survey conducts interviews throughout the year and month of interview is not publicly available. Therefore, the reference period for someone interviewed in 2015, for example, could be any twelve months in the January 2014 to December 2015 range.

income limit, that is, the baseline income limit that applies to adults without children or disabilities (measured as a percentage of the poverty line) in effect in county *c* in state *s* for the majority of year *t*. Using the Medicaid income limit rather than a binary expansion variable has the advantage that we incorporate information on the generosity of Medicaid eligibility prior to Medicaid expansion, thereby exploiting the fact that policy changes in some states were larger than others. The vector X_{ct} includes time-varying controls such as demographic characteristics, and φ_c and τ_t are county and time fixed effects, included to account for unmeasured heterogeneity in outcomes across space and time that may be correlated with expansion status.

This equation corresponds to the standard two-way fixed effects approach used in the ACA Medicaid expansion literature thus far, although it has typically been estimated at the state level or individual level with state and year fixed effects rather than at the county level. As a preliminary step, we estimate equation (1) using the county-level data, clustering standard errors at the state level to account for the fact that the variation in expansion status is at the state level. We weight by county working age population or total population, depending on the outcome, to account for the substantial variation in population size across counties.

To deal with the potential empirical challenge posed by the strong geographic correlation in state expansion decisions together with differential trends in the outcomes of interest across different parts of the country, our preferred specification builds on equation (1). Specifically, it compares changes in program participation within contiguous pairs of counties that cross state lines, where the states experienced differential expansion.[9] Relative to the standard approach, the county border design has the advantage that bordering counties are more likely to share similar labor markets, are more likely to be affected by the same local trends, and are more likely to share macroeconomic shocks than are those that do not share a common border (Allegretto et al. 2013; Dube, Lester, and Reich 2016). The identifying assumption is that the change in the outcome of interest in a county is a reasonable counterfactual estimate for how the outcome of interest would have changed in its neighboring county across the border if the Medicaid expansion had not occurred.

[9] This approach was pioneered in studies of the employment effects of state minimum wages (see Dube, Lester, and Reich 2010, 2016), and we have used it in previous work examining the impact of the ACA Medicaid expansion on disability program applications (Schmidt, Shore-Sheppard, and Watson 2020).

We limit the sample to contiguous border counties and restructure the data so that each county-county pair-year is one observation. Then we estimate a modified version of equation (1):

$$(2) \qquad y_{ct} = \alpha + \delta NoncategoricalLimit_{s(c)t} + X_{ct}\Gamma + \varphi_c + \tau_{pt} + \varepsilon_{cpt}$$

where the subscript p denotes a county-pair and τ_{pt} is a pair-specific time effect (instead of a national time effect). The use of the pair-specific time effect means that we are using only variation in expansion status within each contiguous border county pair and controlling for local economic shocks that impact both members of the county pair in a given year. Because of the data restructuring, some county-years are observed more than once because they border multiple counties. We re-weight, ensuring that the weights across county-county-pair-year observations within a county-year sum to the county population. We cluster standard errors by state and adjust to account for duplicated observations. Because model (2) is only relevant to counties on state borders, and because some county-years are observed more than once, the models in equations (1) and (2) are estimated on different samples. As a check, we estimate the two-way fixed effects model of equation (1) on the restructured subsample of counties used in the estimation of equation (2).

We also use the county border-pair approach for the analysis of the restricted-access ACS, estimating a version of equation (2) on individual-level data. Outcomes for the ACS include individual reports of insurance coverage, program participation, imputed EITC eligibility, and various measures of labor supply on both the extensive and intensive margins. As in the administrative analysis, we examine the effect of a change in the non-categorical Medicaid income limit. We include indicator controls for demographic group (four groups based on parental status, marital status, and age, as described above), non-Hispanic Black interacted with demographic group, Hispanic interacted with demographic group, female, high school education, some college education, and college graduate. We also control for individual year of age. Each coefficient presented in the ACS tables represents the key estimate on the non-categorical income limit from a different regression that includes county-pair by year fixed effects, equivalent to column 3 of the county-level analysis tables. Clustered standard errors are shown in parentheses. Below the clustered standard errors, we show False Discovery Rate q-values to reflect inference with multiple hypothesis tests, as suggested by Andersen (2008). The asterisks on the regression coefficients refer to p-values unadjusted for multiple hypothesis testing, whereas the asterisks on the bracketed q-values refer to the q-value significance levels.

One possible concern with the county-border-pair identification strategy is that individuals might migrate across county lines in order to obtain Medicaid. However, evidence suggests that any such migration is likely to be minimal. The affected households have low incomes and are likely to face barriers to moving in response to the Medicaid expansion such as limited time and resources to search for housing (Bergman et al. 2020). Goodman (2017) finds no evidence of a migration response to the ACA Medicaid expansion at the public-use microdata area (PUMA) level, consistent with findings by Schwartz and Sommers (2014) for earlier health insurance expansions. We also confirm in our ACS work that the sign and significance of most coefficients is unchanged when restricting to a sample of those who did not migrate across state lines.

While the border county discontinuity approach has strong intuitive appeal since it narrows the comparison to an arguably more similar counterfactual, it is important to demonstrate that restricting to bordering county pairs improves the comparison between expansion and non-expansion states. Various methods of examining the validity of border-pair models have been suggested in the literature (see Dube, Lester, and Reich 2010, 2016; Allegretto et al. 2013; and Neumark, Salas, and Wascher 2014). In Appendix Table 1, we show that pre-ACA (2010-11) mean absolute differences in values of our EITC and SNAP outcomes and covariates are smaller for contiguous pairs than they are for pairs formed by matching every other county with each county in the data. This suggests that county pairs that border each other are indeed more similar on observables—and therefore likely more similar on unobservables—than randomly chosen county pairs, engendering more confidence in the border county design than in a standard difference-in-differences design.

To illustrate the strength of the county-pair design in the context of the ACA, we break the counties into two groups: those that ever expanded Medicaid and those that did not, where "ever expanded" is defined as having a Medicaid eligibility limit above zero for non-parents without disabilities at any point between 2010 and 2016. We then run regressions of our dependent variables of interest on the interaction of the "ever expanded" dummy with dummies for each year, controlling for the main effects of each year as well as other controls used in the regression analysis below. The resulting coefficients tell us about the mean difference between "ever expanded" counties and "never expanded" counties in each year in an outcome of interest, relative to the year 2010.

For example, in Panel A of Figure 2, we show the results of this exercise where the non-categorical Medicaid limit is the dependent variable (bars indicate 95 percent confidence intervals). The hollow triangles show the analysis using a standard county two-way fixed effects specification, the solid triangles restrict to border counties, and the solid circles show our preferred specification which controls for border pair by year effects. As expected, regardless of specification, the difference in income limits between expansion and non-expansion counties grows dramatically in 2014 when most counties implemented their Medicaid expansion.

Panel B of Figure 2 repeats this descriptive exercise using the percent uninsured as a dependent variable. In all three specifications, we see a substantial relative decline in uninsurance starting in 2014 in "ever expanded" counties. There is no evidence of differential trends between "ever expanded" and "never expanded" counties in uninsurance prior to expansion in any of the specifications, and the figure illustrates that the border pair specification more precisely isolates the health insurance effects of policy changes.

These figures are meant to provide a descriptive representation of our empirical approach. They do not account for the timing of expansions, nor do they account for differing levels of the non-categorical income limit in effect in different states at different times. We next present results from our full analyses that take these differences into account to examine safety net program participation.

V. Results from County-Level Administrative Data

A. Uninsurance

Table 3 shows the effects of the ACA Medicaid expansions on uninsurance, controlling for the race/ethnicity composition of the population. Column 1 shows the analysis for the all counties sample with county fixed effects and year fixed effects, using the standard difference-in-differences model with a continuous treatment variable shown in equation (1). The results in column 1 suggest that moving from a non-categorical income limit of 0 percent to 100 percent of the federal poverty level reduces uninsurance by 0.6 percentage points. Column 2 uses the same specification but restricts to the subsample of contiguous county pairs; the point estimate is larger in magnitude at -1.3. Column 3 incorporates county-pair by year fixed effects as described in equation (2) above. The county-pair approach suggests a reduction of uninsurance of about 1.6 percentage points for an increase in the income limit from 0 percent to 100 percent. For context,

the typical expansion moved from an income limit of 0 percent to 138 percent of the poverty line. Therefore, a typical expanding county reduced uninsurance by about 2.2 percentage points relative to an adjacent non-expanding county, around 12 percent of the baseline uninsurance level of 18.7 percent.

Columns 4 through 6 of Table 3 repeat the exercise examining the percent of the population with family incomes under 250 percent of the federal poverty level that is uninsured. The preferred specification in column 6, which includes county-pair by year fixed effects, shows that increasing the Medicaid income limit from 0 to 100 percent of the poverty line reduced uninsurance 2.8 percentage points, with a change in the income limit from 0 to 138 percent again implying a 12 percent reduction relative to the baseline of 29.5 percent uninsurance in 2010. These findings corroborate the results of prior research using state difference-in-differences or incorporating within-state variation in prior uninsured rates (for example, Kaestner et al. 2017 and Courtemanche et al. 2017).

B. Earned Income Tax Credit

Our results for estimation of equations (1) and (2) for tax returns with an EITC per 100 working age adults are shown in Columns 1-3 of Table 4. The point estimates are positive and significant in the standard difference-in-differences specifications in column 1 (all counties) and column 2 (contiguous counties), with estimated coefficients that suggest an increase of 0.17 percentage points for an increase in the Medicaid income limit from 0 to 100 percent of the poverty line. Our preferred estimate in column 3 using the county-pair identification strategy is roughly half the magnitude and not statistically different from zero. The point estimate implies an effect size of 0.09 percentage points, suggesting that an increase in the Medicaid income limit from 0 to 138 percent is associated with an increase of 0.12 EITC returns per 100 working-age adults, approximately 1 percent of the baseline level. Overall, our results suggest that the expansion of Medicaid eligibility under the ACA may have led to a small increase in EITC filing, but the preferred specification does not show a statistically significant relationship. The preferred point estimate is about half of the size of that suggested by the standard difference-in-differences approach, which may be biased upward by preexisting differential trends in EITC returns between expansion and nonexpansion counties.

C. Supplemental Nutrition Assistance Program

Next, we turn to our results for the SNAP administrative data (Columns 4-6 of Table 4). As noted earlier, these results rely on a smaller sample because a number of states do not report county-level SNAP data. We find statistically significant and positive estimates for the effect of the Medicaid non-categorical income limit on SNAP participation per 100 people. The coefficient of 0.43 on our preferred specification (column 6) indicates that an increase in the Medicaid non-categorical income limit from 0 to 138 percent of the poverty level led to a statistically significant change of 0.6 additional SNAP participants per 100 people, a 4 percent increase relative to the mean rate of 15 percent SNAP participation. The estimates are about one-quarter of the effect size on uninsurance.

VI. Results from the Restricted-Access American Community Survey

Overall, our county-level analyses show mixed evidence that EITC participation was affected by the ACA Medicaid expansion and consistently suggest that the expansion led to an increase in SNAP participation. In this section, we use individual-level data from the ACS to further investigate the connections between safety net programs. Specifically, we look at joint program participation as well as participation in TANF, which is not available in county-level administrative data. We also examine heterogeneous impacts across groups based on age and family structure because these groups face different Medicaid, SNAP, and EITC eligibility criteria. In addition, we directly examine one potential mechanism, changes in program eligibility through changes in labor supply, and explore how state policy environments might affect spillovers across programs.

A. Health Insurance and Safety Net Participation

We first investigate whether this new dataset and empirical approach yields the same basic findings about the effect of the policy change on insurance that we document in the county-level data. As expected, we find economically meaningful and statistically significant increases in health insurance for all demographic groups in expansion counties relative to those in adjacent non-expansion counties (first row of Table 5).[10]

[10] The results are not directly comparable to those in Table 3 because Table 3 reflects the full population rather than adults age 25 to 64. Nevertheless, the estimate in column 1 of a 1.74 percentage point increase in any health insurance for an increase in the income limit from 0 to

A strength of the ACS dataset is that it allows us to explore heterogenous effects by demographic subgroups and to distinguish between public and private health insurance. Point estimates suggest that the effects on insurance are most pronounced for younger non-parents and single parents (see columns (2) and (4) of Table 5). Insurance effects are larger for the sub-sample under 130 percent of the poverty line. The increases in insurance are driven by increases in public health insurance in expansion counties, partially offset by reductions in private insurance. These estimates of impacts of the expansion on any insurance, public insurance, and private insurance are fairly similar in magnitude to previous estimates in the literature using different identification strategies (Kaestner et al. 2017, Courtemanche et al. 2017).

Turning next to other safety net program participation, the ACS results using the border county design in Table 5 show positive impacts on SNAP participation, consistent with the results from the analysis of county-level administrative data. The coefficient in the full sample analysis suggests a 0.49 percentage point increase in the probability a sample adult lives in a SNAP household. The magnitude is not very different from the 0.43 percentage point increase identified in our analysis of county-level administrative data.

The results in Table 5 broken down by demographic group show that the effect on SNAP holds across most demographic groups (though not statistically significant for young non-parents or married parents) and the point estimate is largest for single parents. The estimated effect on SNAP is larger under 130 percent of the poverty line (the SNAP income eligibility threshold) overall and in most sub-groups, indicating that changes in income eligibility alone are not driving the results.

We also examine multiple program participation. Medicaid eligibility expansion is associated with an increase in those who participate in both Medicaid and SNAP, and a decline in those who participate in SNAP without Medicaid. The numbers imply that about a fifth of those induced to obtain public insurance as the result of ACA expansions (2.50 of every 100 adults for an increase in the income limit of 100 percent of the poverty line) are also induced to participate in SNAP (0.49 of every 100 adults) when they otherwise would not have done so.

---

100 percent of the poverty line is fairly similar to the estimated 1.62 percentage point decline in uninsurance in Table 3.

These numbers are quite similar to what is indirectly implied by the administrative analysis, in which the decline in uninsurance is roughly four times the magnitude of the increase in SNAP.

Despite the fact that income eligibility levels are similar for Medicaid and SNAP, the Medicaid expansion led to an increase in individuals reporting receiving Medicaid but not SNAP of between 0.7 and 1.9 percentage points, depending on the demographic group, suggesting that any increase in information or reduction in administrative burden was not sufficient to enroll all individuals eligible for both programs.

Table 5 also shows coefficients for participation in TANF. These results suggest an increase in TANF participation, particularly among single parents (the group most often eligible for TANF), in expansion counties. The coefficient for single parents implies an increase in participation of 0.49 percentage points (on a base of 6 percent) from a 100 percentage-point increase in the Medicaid income limit. This finding is consistent with new Medicaid enrollees learning about or getting access to other safety net programs. Unfortunately, EITC receipt is not asked about in the ACS, so we are only able to examine mechanisms affecting EITC eligibility due to changes in earned income.[11] We discuss this in the following sub-section.

B. Mechanisms

We next investigate labor supply responses that could explain a link between Medicaid eligibility and participation in other safety net programs. As noted above, Medicaid expansions could affect labor supply decisions in either direction, which in turn may affect eligibility for other programs.

In Table 6, we start by examining imputed EITC eligibility, which could be more prevalent if individuals enter the labor force as the result of the ACA, or if individuals reduce their labor supply so that they fall within the EITC earnings range. The point estimate in column (1) of Table 6 does suggest a small increase in imputed EITC eligibility (0.24 percentage points for a 100 percentage-point increase in the Medicaid income limit), relative to a base of 14 percent. However, the coefficient does not retain significance after adjusting for multiple hypothesis testing. The largest point estimate is for single parents. This finding, though small and

[11] We attempted to examine the information mechanism for EITC participation in our county data using the measure of knowledge about the EITC in the county developed by Chetty, Friedman, and Saez (2013) to examine whether there were differential responses to Medicaid expansion in counties with high versus low levels of knowledge, but the results were statistically inconclusive.

statistically fragile, is consistent with a model of labor supply responses to increased Medicaid income limits among single parents generating spillover effects onto EITC eligibility. As noted below, other evidence in Table 6 points to labor force entry among single parents as the result of ACA expansions, but these findings are sensitive to multiple hypothesis adjustments.

We look at whether Medicaid expansions impact the probability of ACS respondents being under 130 percent of the poverty line, the eligibility threshold for SNAP. There is suggestive evidence of an increase in being under 130 percent shown in Table 6 overall and for some sub-groups, but the statistical significance does not hold up to adjustment for multiple hypothesis testing. There is similarly no statistically robust effect on being in more narrow income groups below 130 percent of the poverty line.

We also look at labor supply by examining a variety of measures of earnings, hours, and business income. Most of these labor supply coefficients are not statistically significant after adjusting for multiple hypothesis testing. One exception is suggestive evidence of an increase in labor supply on the extensive margin among single parents. This is consistent with the possibility that some single parents enter the labor market under the ACA because they no longer lose health insurance coverage when doing so. The labor supply response of single parents to ACA expansions is worthy of further investigation.

Prior literature based on state difference-in-differences models has found limited evidence of a labor supply response to ACA Medicaid expansions (e.g., Gooptu et al. 2016, Kaestner et al. 2017, Frisvold and Jung 2018, Leung and Mas 2018). The overall results of our ACS border county analysis of labor supply measures, including on both the extensive and intensive margins, are consistent with this characterization.

These small estimated labor supply effects suggest that increased information or reduced transaction costs are the primary explanations for increased levels of participation in the SNAP program resulting from the Medicaid expansion. It is possible that changes in labor supply may have contributed to small changes in EITC participation, but we lack necessary statistical precision to draw any firm conclusions. Coupled with the large SNAP changes we see in Table 5 for those who are under the eligibility threshold, we infer that Medicaid expansions mainly induced program participation through an information/transaction costs channel.

C. Policy Environment

We further probe the relationship between Medicaid expansion and SNAP participation by investigating whether baseline policy environments affect the magnitude of the estimated spillovers between the ACA Medicaid expansion and SNAP participation. Specifically, we investigate whether the SNAP response is bigger in states in which (a) there existed a large pool of individuals who were income-eligible for SNAP but were not participating prior to 2014, (b) the SNAP enrollment process has fewer barriers, and (c) there are relaxed SNAP eligibility requirements.

For this analysis, we move to a traditional state-level approach to maximize statistical power.[12] Panel A of Table 7 implements a two-way fixed effects model in the restricted-access American Community Survey sample, for the full sample (Column 1) and for our demographic subgroups (Columns 2-5). We see a statistically significant positive impact of Medicaid expansion on SNAP, similar in magnitude to the county-pair estimates in Table 5. These impacts are statistically significant for the full sample, and for younger and older non-parents.

In Panel B of Table 7, we examine whether the 2013 state and local policy environment affects SNAP spillovers, by interacting the continuous Medicaid income limit with measures of the policy environment. We calculate unmet need or scope for increased enrollment in the SNAP program, using estimates of state SNAP take-up rates in 2013 from Cunnyngham, Sukasih, and Castner (2016) to calculate the estimated fraction of SNAP-eligible people in the state who do *not* participate.

To investigate the role of administrative burden, we generate a measure of SNAP ease of enrollment from components of the SNAP policy index in Ganong and Liebman (2018). States can choose to implement policy options that reduce administrative burden and make it easier for potential recipients to access benefits. The enrollment index combines measures of three state policy choices regarding SNAP: (a) whether states have call centers, (b) whether they allow online applications, and (c) whether Supplemental Security Income recipients have a simplified application process through the Combined Application Project (CAP). The enrollment index ranges from 0 to 3, reflecting the number of the three policy options that are in place as of 2013.

[12] There are an inadequate number of discordant border pairs to make these interactive comparisons using the border pair strategy feasible.

Finally, we examine differential impacts based on an index of SNAP eligibility, which is higher in states with more generous eligibility rules. We consider the 2013 presence of two policies: (a) a waiver of the work requirement for able-bodied adults without dependents (ABAWD) in the majority of the county, making it easier for adults without dependents to qualify, and (b) broad-based categorical eligibility (BBCE) for SNAP in the state. BBCE relaxes some income and asset requirements for low-income individuals who participate in non-cash TANF programs.[13] The SNAP eligibility index is equal to 2 if both policies are in place, 1 if just one of these policies is in place, and 0 if neither.

The results in Panel B suggest that some elements of the policy environment lead to larger spillovers between the Medicaid expansion and SNAP, and this is true both for the full sample and all demographic subgroups. States with more scope to enroll eligible people in SNAP ("unmet need") did so to a greater extent as a result of Medicaid expansion. The coefficient suggests that a fully expanding state with a lot of unmet need, like Nevada with 2013 SNAP take-up of 66 percent, would see 1.2 percentage points more of a boost in SNAP participation than a place like Colorado with a baseline take-up rate of 84 percent. It is likely that ACA expansions encouraged some of those who were previously disconnected from safety net programs to enroll in multiple programs at once.

In addition, states with more relaxed SNAP eligibility rules indeed had more SNAP take-up as a result of ACA expansions, although we cannot rule out the possibility that other state factors correlated with SNAP eligibility rules, including state attitudes towards participation, might be driving this result. We find no significant effects of the ease of SNAP enrollment variable on spillovers from the Medicaid expansion to SNAP. This absence of a significant relationship does not necessarily imply that administrative burden is an unimportant determinant of safety net take-up, but rather that these particular measures of administrative burden are not strong predictors of cross-program spillovers, perhaps because they do not adequately capture heterogeneity in the marginal cost of applying to SNAP at the time of Medicaid enrollment.

[13] Data on broad-based categorical eligibility and most other state policy options are provided by online data accompanying Ganong and Liebman (2018) and are based on SNAP state policy option reports.

VII. Robustness

We perform several robustness checks (not shown). County-level results are robust to inclusion of additional controls for the age distribution of the population, to the inclusion of the (potentially endogenous) unemployment rate, and to the use of a binary expansion variable instead of our continuous non-categorical Medicaid limit. We consider a county to have expanded in a given year if it has a non-categorical Medicaid limit greater than zero.

In the ACS analysis, we address concerns about potential endogenous migration by restricting to a subsample who did not live in a different state in the prior year. For the most part, coefficients maintain their sign and significance when restricting the samples in this way.

Finally, while the majority of our states expanded Medicaid in 2014, we examine whether the known issues with two-way fixed effects models with staggered timing might be affecting the results.[14] We switch to a simple county-level two-way fixed effects analysis and a binary expansion variable to investigate the role of staggered timing.[15] We decompose our estimates and show the weight that each type of "treatment" versus "control" comparison has in the overall estimate. The decomposition suggests that the majority (82 percent) of our estimate is driven by four comparisons: never-treated to 2012 expanders, never-treated to 2014 expanders, never-treated to 2015 expanders, and 2014 expanders to 2012 expanders. The difference-in-difference estimates on the first three are all positive, while the estimate on the last (and the only invalid comparison in the presence of dynamic treatment effects) is negative. If we manually take out the invalid comparisons and reweight our estimates, the effect of the ACA Medicaid expansion on SNAP participation becomes even larger in magnitude.

As an additional robustness test, we estimate county-level regressions comparing all states that expanded in a given year with those that never expanded (i.e., compare states that expanded in 2011 with states that never expanded, compare states that expanded in 2012 with those that never expanded, etc., separately for all expansion years through 2016). Effects of the

[14] Other papers show that the staggered timing of the Medicaid expansion does not appear to create large biases for their outcomes (see, for example, Carey, Miller, and Wherry (2020) on utilization of health care among Medicare beneficiaries).

[15] Specifically, we use a two-way fixed effects model with a binary treatment variable representing expansion, defined as any non-categorical Medicaid eligibility for adults. The sample counties are restricted to be a balanced panel and to eliminate one state which was not monotonically increasing in treatment status (Goodman-Bacon, Goldring, and Nichols 2019).

Medicaid expansion on SNAP participation from these regressions are positive in all years, though they vary in statistical significance.

VIII. Conclusion

Our results suggest that the Medicaid expansion affected safety net program participation in counties in states that expanded relative to nearby counties in states that did not expand. In both county-level administrative data and the restricted-access American Community Survey, we find robust evidence that the Medicaid expansion increased SNAP take-up, with about a fifth to a quarter of those gaining public insurance due to the Medicaid expansion also induced to participate in SNAP. We also find evidence of increases in TANF receipt in the ACS. However, while administrative data at the county and state levels show suggestive evidence of small increases in EITC receipt, these increases are not statistically significant in the preferred border-pair specifications.

Our analysis of mechanisms using the American Community Survey shows no strong evidence of changes in eligibility due to earnings or income, but instead points to information costs or other administrative burdens as the main explanation for the observed association between Medicaid and SNAP. It is likely that the process of enrolling in Medicaid facilitates SNAP enrollment. These spillovers were most pronounced in places with more flexible SNAP eligibility rules and in places with low SNAP take-up in 2013, suggesting that the ACA expansion helped to connect disconnected individuals with unmet need to the safety net.

The results have implications for the large body of literature examining the effect of Medicaid expansions. It is possible that some of the benefits of the expansion occurred because families were connected to food or cash assistance programs rather than solely as a direct result of health insurance provision. In other words, safety net spillovers may be an important mechanism for findings in the literature.

The conclusion that Medicaid expansions promote SNAP, TANF, and perhaps EITC participation suggests the federal costs associated with the ACA are greater than the direct costs of the program itself. Similarly, states considering Medicaid expansions might consider additional inflows of federal SNAP and EITC dollars as a benefit of making Medicaid income eligibility limits more generous. More broadly, the findings suggest the presence of important spillover effects across safety net programs.

IX. References

Allegretto, Sylvia, Arindrajit Dube, Michael Reich, and Ben Zipperer. 2013. "Credible Research Designs for Minimum Wage Studies." IZA Discussion Paper No. 7638.

Andersen, Michael L. 2008. "Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects." *Journal of the American Statistical Association* 103(484): 1481-1495.

Baicker, Katherine, Amy Finkelstein, Jae Song, and Sarah Taubman. 2014. "The Impact of Medicaid on Labor Market Activity and Program Participation: Evidence from the Oregon Health Insurance Experiment." *American Economic Review* 104(5):322-328.

Barcellos, Silvia H, Mireille Jacobson, and Helen G. Levy. 2023. "The Impact of Eligibility for Medicaid versus Subsidized Private Health Insurance on Medical Spending, Self-Reported Health, and Public Program Participation." *American Journal of Health Economics* 9(2): 262-295.

Bergman, Peter, Raj Chetty, Stefanie DeLuca, Nathaniel Hendren, Lawrence F. Katz, and Christopher Palmer. 2020. "Creating Moves to Opportunity: Experimental Evidence on Barriers to Neighborhood Choice." National Bureau of Economic Research Working Paper 26164.

Burney, Shaheer, Rebecca Boehm, and Rigoberto Lopez. 2021. "The Impact of the ACA Medicaid Expansion on SNAP Participation." *Food Policy* 101(May).

Brooks, Tricia, Joe Touschner, Samantha Artiga, Jessica Stephens, and Alexandra Gates. 2015. "Modern Era Medicaid: Findings from a 50-State Survey of Eligibility, Enrollment, Renewal, and Cost-Sharing Policies in Medicaid and CHIP as of January 2015." Kaiser Commission on Medicaid and the Uninsured Policy Brief #8681.

Brooks, Tricia, Sean Miskell, Samantha Artiga, Elizabeth Cornachione, and Alexandra Gates. 2016. "Medicaid and CHIP Eligibility, Enrollment, Renewal, and Cost-Sharing Policies as of January 2016: Findings from a 50-State Survey." Kaiser Commission on Medicaid and the Uninsured Policy Brief #8824.

Carey, Colleen M., Sarah Miller, and Laura R. Wherry. 2020. "The Impact of Insurance Expansion on the Already Insured: The Affordable Care Act and Medicare." *American Economic Journal: Applied Economics* 12(4): 288-318.

Cha, Paulette, and José J. Escarce. 2022. "The Affordable Care Act Medicaid Expansion: A Difference-in-Differences Study of Spillover Participation in SNAP." *PloS one* 17, no. 5 : e0267244.

Chetty, Raj, John Friedman and Emmanuel Saez. 2013. "Using Differences in Knowledge Across Neighborhoods to Uncover the Impacts of the EITC on Earnings," *American Economic Review* 103(7): 2683-2721.

Cohen Ross, Donna, Marian Jarlenski, Samantha Artiga, and Caryn Marks. 2009. "A Foundation for Health Reform: Findings of a 50 State Survey of Eligibility Rules, Enrollment and Renewal Procedures, and Cost-Sharing Practices in Medicaid and CHIP for Children and Parents During 2009." Kaiser Commission on Medicaid and the Uninsured Policy Brief #8028.

Courtemanche, Charles, James Marton, Benjamin Ukert, Aaron Yelowitz, and Daniela Zapata. 2017. "Early Impacts of the Affordable Care Act on Health Insurance Coverage in Medicaid Expansion and Non-Expansion States." *Journal of Policy Analysis and Management* 36(1): 178–210.

Cunnyngham, Karen E. 2012. "Reaching Those in Need: State Supplemental Nutrition Assistance Program Participation Rates in 2010." Alexandria, VA: U.S. Department of Agriculture, Food and Nutrition Service, February. https://fns-prod.azureedge.net/sites/default/files/ops/Reaching2010.pdf

Cunnyngham, Karen, Amang Sukasih, and Laura Castner. 2016. "Empirical Bayes Shrinkage Estimates of State Supplemental Nutrition Assistance Program Participation Rates in Fiscal Year 2011 to Fiscal Year 2013 for All Eligible People and the Working Poor." Mathematica Policy Research, Report to the U.S. Department of Agriculture. https://www.mathematica.org/publications/empirical-bayes-shrinkage-estimates-of-snap-participation-rates-in-fiscal-year-2011-to-fiscal-year.

Dube, Arindrajit, T. William Lester, and Michael Reich. 2010. "Minimum Wage Effects Across State Borders: Estimates Using Contiguous Counties." *Review of Economics and Statistics* 92(4): 945-964.

Dube, Arindrajit, T. William Lester and Michael Reich. 2016. "Minimum Wage Shocks, Employment Flows, and Labor Market Frictions." *Journal of Labor Economics* 34(3): 663-704.

Duggan, Mark, Gopi Shah Goda, and Emilie Jackson. 2019. "The effects of the Affordable Care Act on health insurance coverage and labor market outcomes." *National Tax Journal* 72(2): 261-322.

Falk, Gene. 2014. "Temporary Assistance for Needy Families (TANF): Eligibility and Benefit Amounts in State TANF Cash Assistance Programs." Congressional Research Service Report, July 22.

Finkelstein, Amy and Matthew J. Notowidigdo. 2019. "Take-Up and Targeting: Experimental Evidence from SNAP." *Quarterly Journal of Economics* 134(3): 1505-1556.

Frisvold, David E. and Younsoo Jung. 2018. "The impact of expanding Medicaid on health insurance coverage and labor market outcomes." *International Journal of Health Economics and Management* 18 (2): 99-121.

Ganong, Peter and Jeffrey B. Liebman. 2018. "The Decline, Rebound, and Further Rise in SNAP

Enrollment: Disentangling Business Cycle Fluctuations and Policy Changes." *American Economic Journal: Economic Policy* 10(4): 153–176.

Goodman, Lucas. 2017. "The Effect of the Affordable Care Act Medicaid Expansion on Migration." *Journal of Policy Analysis and Management* 36(1): 211-238.

Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225(2): 254-277.

Goodman-Bacon, Andrew, Thomas Goldring, and Austin Nichols. 2019. "BACONDECOMP: Stata module to perform a Bacon decomposition of difference-in-differences estimation," Statistical Software Components S458676, Boston College Department of Economics, revised 15 Sep 2019.

Gooptu, Angshuman, Asako S. Moriya, Kosali I. Simon, and Benjamin D. Sommers. 2016. "Medicaid Expansion Did Not Result In Significant Employment Changes Or Job Reductions In 2014." *Health Affairs* 35 (1): 111-118.

Heberlein, Martha, Tricia Brooks, Jocelyn Guyer, Samantha Artiga, and Jessica Stephens. 2011. "Holding Steady, Looking Ahead: Annual Findings of a 50-State Survey of Eligibility Rules, Enrollment and Renewal Procedures, and Cost-Sharing Practices in Medicaid and CHIP, 2010-2011." Kaiser Commission on Medicaid and the Uninsured Policy Brief #8130.

Heberlein, Martha, Tricia Brooks, Jocelyn Guyer, Samantha Artiga, and Jessica Stephens. 2012. "Performing Under Pressure: Annual Findings of a 50-State Survey of Eligibility, Enrollment, Renewal, and Cost-Sharing Policies in Medicaid and CHIP, 2011-2012." Kaiser Commission on Medicaid and the Uninsured Policy Brief #8272.

Heberlein, Martha, Tricia Brooks, Joan Alker, Samantha Artiga and Jessica Stephens. 2013. "Getting into Gear for 2014: Findings from a 50-State Survey of Eligibility, Enrollment, Renewal, and Cost-Sharing Policies in Medicaid and CHIP, 2012–2013." Kaiser Commission on Medicaid and the Uninsured Policy Brief #8401.

Herd, Pamela and Donald P. Moynihan. 2018. *Administrative Burden: Policymaking by Other Means*, New York: Russell Sage Foundation.

Herd, Pamela, Hilary Hoynes, Jamila Michener, and Donald Moynihan. 2023. "Introduction: Administrative Burden as a Mechanism of Inequality in Policy Implementation." RSF: The Russell Sage Foundation Journal of the Social Sciences.
Hoynes, Hilary and Diane Whitmore Schanzenbach. 2016. "US Food and Nutrition Programs." In *Economics of Means-Tested Transfer Programs in the United States, Volume 1*, Robert Moffitt, ed., Chicago: University of Chicago Press, pp. 219-301.

Internal Revenue Service. 2013. "Revenue Procedure 2013-35." https://www.irs.gov/pub/irs-drop/rp-13-35.pdf.

Internal Revenue Service. 2019. "EITC Awareness Day." https://www.eitc.irs.gov/partner-toolkit/eitc-awareness-day/eitc-awareness-day-2

Jones, Maggie R. 2014. "Changes in EITC Eligibility and Participation, 2005–2009." CARRA Working Paper #2014-04. Washington, DC: US Census Bureau, July. https://www.census.gov/content/dam/Census/library/working-papers/2014/adrm/carra-wp-2014-04.pdf.

Kaestner, Robert, Bowen Garrett, Jiajia Chen, Anuj Gangopadhyaya, and Caitlyn Fleming. 2017. "Effects of ACA Medicaid expansions on health insurance coverage and labor supply." *Journal of Policy Analysis and Management* 36(3): 608-642.

Kaiser Commission on Medicaid and the Uninsured. 2013. "A Look at Section 1115 Medicaid Demonstration Waivers Under the ACA: A Focus on Childless Adults." Issue Brief. https://kaiserfamilyfoundation.files.wordpress.com/2013/10/8499-a-look-at-section-1115-medicaid-demonstration-waivers.pdf

Leung, Pauline, and Alexandre Mas. 2018. "Employment Effects of the Affordable Care Act Medicaid Expansions." *Industrial Relations: A Journal of Economy and Society* 57 (2): 206-234.

National Conference of State Legislatures. 2018. "Tax Credits for Working Families: Earned Income Tax Credit (EITC)." http://www.ncsl.org/research/labor-and-employment/earned-income-tax-credits-for-working-families.aspx

Neumark, David, J. M. Ian Salas, and William Wascher. 2014. "Revisiting the Minimum Wage-Employment Debate: Throwing Out the Baby with the Bathwater?" *Industrial and Labor Relations Review* 67(Supplement):608-648.

Nichols, Austin and Jesse Rothstein. 2016. "The Earned Income Tax Credit." In *Economics of Means-Tested Transfer Programs in the United States, Volume 1*, Robert Moffitt, ed., Chicago: University of Chicago Press, pp. 137-218.

Pavetti, LaDonna, Ali Safawi, and Danilo Trisi. 2021. "TANF at 25: A Weaker Cash Safety Net Reaching Fewer Families and doing Less to Lift Families out of Deep Poverty. *National Tax Journal* 74(3): 763-789.

Schmidt, Lucie, Lara Shore-Sheppard, and Tara Watson. 2020. "The Impact of the ACA Medicaid Expansion on Disability Program Applications." *American Journal of Health Economics* 6(4): 444-476.

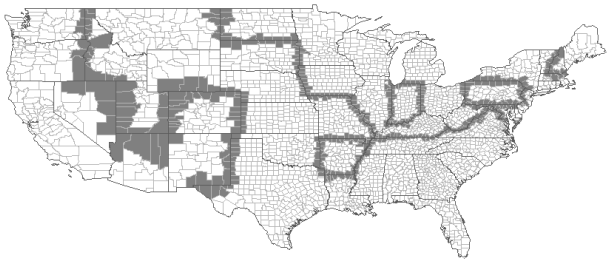
Schwartz, Aaron and Benjamin D. Sommers. 2014. "Moving for Medicaid? Recent Eligibility Expansions Did Not Induce Migration From Other States." *Health Affairs* 33(1):88-94.

Soni, Aparna, Laura R. Wherry, and Kosali I. Simon. 2020. "How Have ACA Insurance Expansions Affected Health Outcomes? Findings from The Literature: A literature review of the

Affordable Care Act's effects on health outcomes for non-elderly adults." *Health Affairs* 39(3): 371-378.

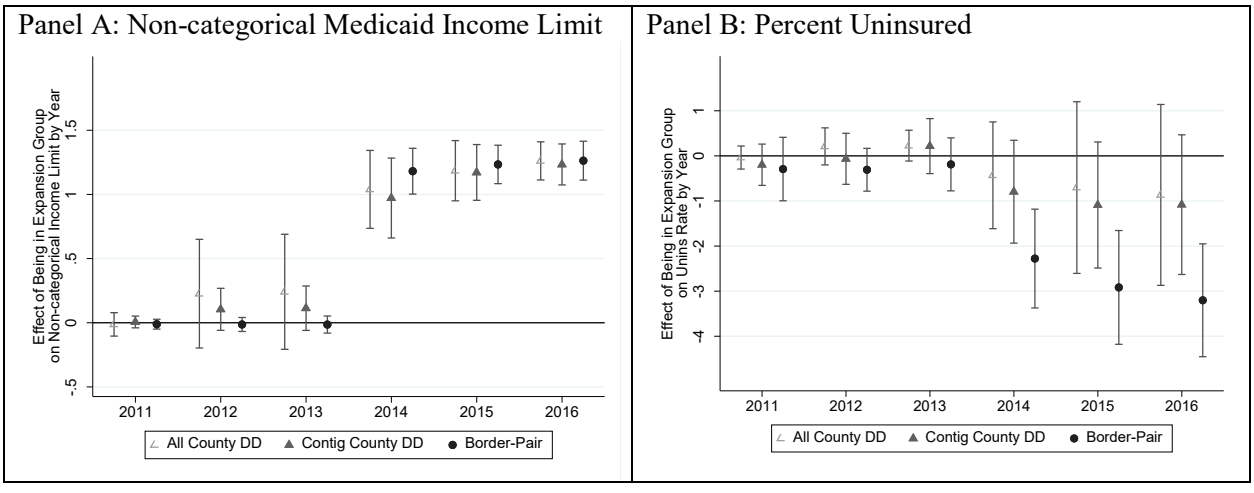
Ziliak, James P. 2016. “Temporary Assistance for Needy Families.” In *Economics of Means-Tested Transfer Programs in the United States, Volume 1*, Robert Moffitt, ed., Chicago: University of Chicago Press, pp. 303-393.

Figure 1: Contiguous Border County Pairs in the US with a Medicaid Expansion Differential, April 2014



Source: Medicaid expansion status determined from data on state actions collected by the Kaiser Family Foundation.

Figure 2. Descriptive Exercise: Effect of Expansion using Different Empirical Approaches



Notes: Coefficients from regressions on 2010-2016 data that include county fixed effects and controls for fraction Black and fraction Hispanic. The "Contig County DD" restricts to counties included in the border-pair analysis, and the "Border-Pair" includes border pair by year fixed effects.

Table 1: Weighted Means by Sub-Sample and Year for County-Level Analysis

Sample and Year	All Counties EITC Sample	Border Counties EITC Sample	All Counties SNAP Sample	Border Counties SNAP Sample
Panel A: 2010 Data				
EITC Returns per 100 Working Age Adults or SNAP Participation Rate	12.959	12.551	14.170	14.565
Fraction Uninsured	17.685	15.974	18.673	17.268
Fraction Low-Income Uninsured	28.257	26.232	29.528	27.741
ACA Medicaid Expansion (Y/N)	0.125	0.157	0.032	0.032
Medicaid Non-Categorical Income Limit (Relative to the Poverty Line)	0.128	0.62	0.038	0.043
Percent Non-Hispanic Black	13.278	13.734	14.167	15.649
Percent Hispanic	16.454	12.750	17.640	13.794
Observations	*3089*	*2390*	*2466*	*1781*
Unique counties	*3089*	*1138*	*2466*	*859*
Panel B: 2016 Data				
EITC Returns per 100 Working Age Adults or SNAP Participation Rate	14.235	13.919	13.254	13.929
Fraction Uninsured	9.965	8.903	10.675	9.699
Fraction Low-Income Uninsured	15.960	14.613	16.947	15.592
ACA Medicaid Expansion (Y/N)	0.618	0.694	0.577	0.673
Medicaid Non-Categorical Income Limit (Relative to the Poverty Line)	0.902	1.005	0.804	0.944
Percent Non-Hispanic Black	13.163	14.076	14.535	16.008
Percent Hispanic	18.000	14.249	19.169	15.592
Observations	*3089*	*2390*	*2399*	*1713*
Unique Counties	*3089*	*1138*	*2399*	*833*

Notes: For the overall EITC sample, EITC returns per 100 range from 0 to 51.6 with a mean of 14.2. For the SNAP sample, the SNAP participation rate ranges from 0 to 154.6 with a mean of 14.4.

Table 2. Means for Individual-Level ACS Analysis

	(1) All (N=11.7m)	(2) Non-Parents<50 (N=2.96m)	(3) Non-Parents 50-64 (N=4.58m)	(4) Single Parents (N=0.964m)	(5) Married Parents (N=3.23m)
Any Health Insurance	0.879	0.819	0.904	0.820	0.935
Any Health Insurance <130 percent of Poverty	0.727	0.631	0.765	0.761	0.780
Public Health Insurance	0.169	0.147	0.193	0.327	0.102
Private Health Insurance	0.748	0.697	0.764	0.525	0.866
Any SNAP	0.138	0.135	0.116	0.376	0.080
Any SNAP < 130 Percent of Poverty	0.494	0.428	0.434	0.654	0.499
SNAP & Public Health Ins	0.077	0.063	0.069	0.232	0.045
SNAP & No Public Health Ins	0.061	0.072	0.047	0.144	0.035
No SNAP & Public Health Ins	0.092	0.084	0.124	0.095	0.057
Any TANF	0.018	0.018	0.014	0.060	0.008
Imputed EITC Eligibility	0.136	0.116	0.048	0.424	0.167
Income < 130 Percent of Poverty	0.218	0.251	0.173	0.488	0.141
Income < 50 Percent of Poverty	0.087	0.122	0.063	0.226	0.029
Any Earnings	0.791	0.838	0.699	0.806	0.854
Any Usual Hours Worked	0.792	0.839	0.700	0.806	0.855
Full-time Hours	0.657	0.707	0.571	0.635	0.720
Any Earner (Self or Spouse)	0.864	0.860	0.790	0.806	0.984
Any Wages (Self or Spouse)	0.833	0.832	0.752	0.771	0.962
Number of Earners	1.228	1.065	1.117	0.806	1.706
Any Business Income (Self or Spouse)	0.098	0.067	0.109	0.049	0.136
Log (Combined Earnings)	10.85	10.54	10.87	10.10	11.35

Notes: Sample is individual citizens living in border counties ages 25 to 64 from the 2011-2017 American Community Survey. Sample means are for the reshaped border county sample used in analyses for Tables 5 and 6.

Table 3: Effects of ACA Medicaid Income Limits on Percent Uninsured and Low Income Uninsured

	(1) All Counties Sample	(2) Contig Counties Sample	(3) Contig Counties Sample	(4) All Counties Sample	(5) Contig Counties Sample	(6) Contig Counties Sample
	Percent Uninsured	Percent Uninsured	Percent Uninsured	Percent Low-Income Uninsured	Percent Low-Income Uninsured	Percent Low-Income Uninsured
Non-Categorical	-0.566**	-1.270***	-1.620***	-1.475***	-2.382***	-2.778***
Income Limit	(0.277)	(0.446)	(0.326)	(0.454)	(0.608)	(0.591)
Percent Non-	0.889*	0.631*	0.341***	0.476	0.109	-0.390
Hispanic Black	(0.451)	(0.365)	(0.127)	(0.666)	(0.439)	(0.282)
Percent Hispanic	-0.441	-0.154	-0.058	-0.319	-0.225	0.317
	(0.378)	(0.748)	(0.269)	(0.522)	(0.861)	(0.507)
Observations	21,623	16,730	16,730	21,623	16,730	16,730
R-squared	0.959	0.952	0.993	0.955	0.952	0.989
County FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	NO	YES	YES	NO
County Pair*Year FE	NO	NO	YES	NO	NO	YES

Notes: Dependent variable is the percent of individuals overall or percent of individuals with family incomes under 250 percent of poverty uninsured in a county from the 2010-2016 Small Area Health Insurance Estimates. EITC analysis sample is used and excludes Republic County, Kansas. Robust standard errors clustered on state in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 4: Effects of ACA Medicaid Income Limits on EITC and SNAP

	(1) All Counties Sample EITC	(2) Contig Counties Sample EITC	(3) Contig Counties Sample EITC	(4) All Counties Sample SNAP	(5) Contig Counties Sample SNAP	(6) Contig Counties Sample SNAP
Non-Categorical	0.168**	0.171***	0.088	0.378	0.510*	0.433**
Income Limit	(0.081)	(0.051)	(0.053)	(0.255)	(0.282)	(0.189)
Percent Non-	0.088	0.216***	0.266***	0.006	0.471**	0.776***
Hispanic Black	(0.107)	(0.080)	(0.062)	(0.216)	(0.204)	(0.230)
Percent Hispanic	0.163***	0.163*	0.154*	0.665***	1.030***	0.453
	(0.053)	(0.084)	(0.084)	(0.181)	(0.220)	(0.372)
Observations	21,623	16,730	16,730	16,965	12,197	12,197
R-squared	0.990	0.993	0.998	0.948	0.962	0.990
County FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	NO	YES	YES	NO
County Pair*Year FE	NO	NO	YES	NO	NO	YES

Notes: Dependent variable in columns 1-3 is the number of EITC returns*100 in a county divided by number of working-age adults (ages 20-64) in the county. Republic County, Kansas does not appear in the Statistics of Income county data. EITC estimates are weighted by working-age population. Dependent variable in columns 4-6 is the number of SNAP participants*100 in a county divided by county population. A number of states do not report county-level SNAP data and are excluded from the sample; see text for details. SNAP estimates are weighted by population. Robust standard errors clustered on state in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 5. Health Insurance and Safety Net Outcomes in the ACS

	(1) All (N=11.7m)	(2) Non-Parents<50 (N=2.96m)	(3) Non-Parents 50+ (N=4.58m)	(4) Single Parents (N=0.964m)	(5) Married Parents (N=3.23m)
Any Health Ins	0.0174***	0.0286***	0.0147***	0.0237***	0.0076***
	(0.002)	(0.005)	(0.002)	(0.005)	(0.002)
	<0.001>***	<0.001>***	<0.001>***	<0.001>***	<0.003>***
Any Health Ins Sample<130 Percent of Poverty	0.0595***	0.0869***	0.0577***	0.0369***	0.0535***
	(0.007)	(0.011)	(0.006)	(0.008)	(0.016)
	<0.001>***	<0.001>***	<0.001>***	<0.001>***	<0.001>***
Public Health Ins	0.0250***	0.0336***	0.0212***	0.0342***	0.0171***
	(0.003)	(0.005)	(0.002)	(0.007)	(0.004)
	<0.001>***	<0.001>***	<0.001>***	<0.001>***	<0.001>***
Private Health Ins	-0.0065***	-0.0041	-0.0059***	-0.0104*	-0.0075***
	(0.002)	(0.003)	(0.002)	(0.005)	(0.003)
	<0.002>***	<0.058>*	<0.003>***	<0.023>**	<0.005>***
Any SNAP	0.0049***	0.0046	0.0042***	0.0135***	0.0024
	(0.002)	(0.003)	(0.001)	(0.005)	(0.002)
	<0.006>***	<0.049>**	<0.003>***	<0.006>***	<0.058>*
Any SNAP Sample <130 Percent of Poverty	0.0116**	0.0111	0.0030	0.0174***	0.0102
	(0.005)	(0.007)	(0.006)	(0.006)	(0.010)
	<0.014>**	<0.046>**	<0.149>	<0.004>***	<0.082>*
SNAP & Public Health Ins	0.0120***	0.0144***	0.0090***	0.0265***	0.0076***
	(0.002)	(0.003)	(0.002)	(0.005)	(0.002)
	<0.001>***	<0.001>***	<0.001>***	<0.001>***	<0.002>***

SNAP & No Public Health Ins	-0.0071*** (0.002) <0.001>***	-0.0098*** (0.002) <0.001>***	-0.0048*** (0.001) <0.001>***	-0.0130** (0.005) <0.009>***	-0.0052*** (0.002) <0.004>***
No SNAP & Public Health Ins	0.0130*** (0.002) <0.001>***	0.0192*** (0.002) <0.001>***	0.0123*** (0.002) <0.001>***	0.0077* (0.004) <0.023>**	0.0095*** (0.002) <0.001>***
Any TANF	0.0007 (0.001) <0.082>*	0.0010 (0.001) <0.085>*	-0.0001 (0.001) <0.203>	0.0049* (0.003) <0.023>**	-0.0001 (0.000) <0.203>

Notes: Sample is individual citizens ages 25 to 64 in contiguous county pairs from the 2011-2017 American Community Survey. Each cell is a coefficient on the state non-categorical income limit for the year prior to survey year from a separate regression. Analyses include county fixed effects, county-pair-year fixed effects, and controls for demographic group interacted with Black, demographic group interacted with Hispanic, female, high school education, some college education, college grad, and individual age dummies. Parental status defined by whether a parent reports a child in the household under 19 is their own biological, step, or adoptive child or that of their partner. Robust standard errors clustered on state in parentheses. For coefficients, asterisks reflect p-values associated with the robust standard errors. *** p<0.01, ** p<0.05, * p<0.1. False discovery rate q-values and associated asterisks are shown in brackets. *** q<0.01, ** q<0.05, * q<0.1.

Table 6. Income and Labor Market Outcomes in the ACS

	(1) All (N=11.7m)	(2) Non-Parents <50 (N=2.96m)	(3) Non-Parents 50+ (N=4.58m)	(4) Single Parents (N=0.964m)	(5) Married Parents (N=3.23m)
Imputed EITC Eligibility	0.0024***	0.0035**	0.0009	0.0052*	0.0006
	(0.001)	(0.001)	(0.001)	(0.003)	(0.002)
	<0.111>	<0.161>	<0.442>	<0.213>	<0.820>
Under 130 Percent of Poverty	0.0022*	0.0036	0.0009	-0.0068*	0.0045*
	(0.001)	(0.002)	(0.001)	(0.003)	(0.002)
	<0.277>	<0.294>	<0.615>	<0.213>	<0.213>
Under 50 Percent of Poverty	0.0010	0.0030	0.0007	-0.0044	0.0008
	(0.001)	(0.002)	(0.001)	(0.004)	(0.001)
	<0.437>	<0.291>	<0.542>	<0.437>	<0.565>
Between 50 and 130 Percent	0.0013	0.0006	0.0002	-0.0023	0.0037**
	(0.001)	(0.002)	(0.001)	(0.004)	(0.002)
	<0.291>	<0.820>	<0.851>	<0.599>	<0.213>
Any Earnings	0.0027**	0.0017	0.0034	0.0095***	-0.0001
	(0.001)	(0.002)	(0.002)	(0.003)	(0.002)
	<0.161>	<0.442>	<0.291>	<0.079>*	<0.893>
Any Usual Hours Worked	0.0027**	0.0018	0.0032	0.0100***	0.0000
	(0.001)	(0.002)	(0.002)	(0.003)	(0.002)
	<0.161>	<0.437>	<0.316>	<0.077>*	<0.901>
Full-time Hours	0.0008	0.0028	0.0016	0.0047	-0.0039**
	(0.001)	(0.002)	(0.002)	(0.004)	(0.002)

	<0.587>	<0.317>	<0.553>	<0.437>	<0.213>
Any Earner (Self and/or Spouse)	0.0018	0.0002	0.0022	n/a	-0.0001
	(0.001)	(0.002)	(0.002)		(0.001)
	<0.291>	<0.851>	<0.437>		<0.851>
Any Wages (Self and/or Spouse)	0.0015	-0.0011	0.0014	0.0116***	0.0008
	(0.001)	(0.002)	(0.002)	(0.003)	(0.001)
	<0.437>	<0.624>	<0.599>	<0.022>**	<0.673>
Any Business Income (Self or Spouse)	-0.0003	0.0010	-0.0010	0.0010	-0.0011
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)
	<0.820>	<0.437>	<0.599>	<0.565>	<0.553>
Number of Earners	0.0023	-0.0065**	0.0087**	N/A	-0.0001
	(0.002)	(0.003)	(0.004)		(0.004)
	<0.437>	<0.161>	<0.161>		<0.901>
Log (Combined Earnings)	-0.0087**	-0.0114	-0.0080*	-0.0060	-0.0079
	(0.003)	(0.007)	(0.004)	(0.008)	(0.005)
	<0.161>	<0.291>	<0.250>	<0.553>	<0.308>

Notes: Sample is individual citizens ages 25 to 64 in contiguous county pairs from the 2011-2017 American Community Survey. Each cell is a coefficient on the state non-categorical income limit for the year prior to survey year from a separate regression. Analyses include county fixed effects, county-pair-year fixed effects, and controls for demographic group interacted with Black, demographic group interacted with Hispanic, female, high school education, some college education, college grad, and age dummies. Parental status defined by whether a parent reports a child in the household under 19 is their own biological, step, or adoptive child or that of their partner. Robust standard errors clustered on state in parentheses. For coefficients, asterisks reflect p-values associated with the robust standard errors. *** p<0.01, ** p<0.05, * p<0.1. False discovery rate q-values and associated asterisks are shown in brackets. *** q<0.01, ** q<0.05, * q<0.1.

Table 7. Effect of State Policy Environments on SNAP Spillovers

Dependent Variable: Any SNAP	(1) Full Sample (N=16.56m)	(2) Non-Parents <50 (N=4.26m)	(3) Non-Parents 50+ (N=6.38m)	(4) Single Parents (N=1.34m)	(5) Married Parents (N=4.57m)
Panel A: Non-categorical Income Limit Only					
Non-categorical Income Limit	0.0045**	0.0057**	0.0055***	0.0024	0.0023
	(0.002)	(0.002)	(0.001)	(0.006)	(0.002)
Panel B: Income Limit with Interactions					
Income Limit X Unmet Need	0.0483***	0.0579***	0.0249**	0.1215***	0.0454***
	(0.013)	(0.015)	(0.009)	(0.031)	(0.012)
Income Limit X Enrollment Index	0.0006	0.0003	0.0008	0.0014	0.0004
	(0.002)	(0.003)	(0.002)	(0.004)	(0.002)
Income Limit X Eligibility Index	0.0105***	0.0124***	0.0075***	0.0258***	0.0070**
	(0.003)	(0.004)	(0.002)	(0.006)	(0.003)

Notes: Sample is individual citizens ages 25 to 64 from the 2011-2017 American Community Survey. Each cell is a coefficient on the state non-categorical income limit for the year prior to survey year from a separate regression. Unlike Tables 6 and 7, this analysis uses a two-way fixed effects specification without relying on border counties. Analyses include county fixed effects, year fixed effects, and controls for demographic group interacted with Black, demographic group interacted with Hispanic, female, age, age squared, high school education, some college education, and college grad. Parental status defined by whether a parent reports a child in the household under 19 is their own biological, step, or adoptive child or that of their partner. Panel B also includes the main effect of a within-county demeaned income limit. The three interacted indices are demeaned within county. Robust standard errors clustered on state in parentheses. For coefficients, asterisks reflect p-values associated with the robust standard errors. *** p<0.01, ** p<0.05, * p<0.1.

Appendix Table 1: Mean Absolute Difference Between Counties in All Pairs and Contiguous Border Pairs Pre-ACA, EITC Sample

Mean Absolute Difference Between Counties	All Pairs	Contiguous Border Pairs	Difference
EITC Returns per 100 Working Age Adults	5.343***	3.124***	-2.219***
	(0.101)	(0.099)	(0.091)
Non-Categorical Income Limit	0.100***	0.091***	-0.009
	(0.010)	(0.013)	(0.009)
Unemployment Rate	3.465***	1.881***	-1.585***
	(0.046)	(0.048)	(0.058)
Percent in Poverty	6.852***	4.287***	-2.566***
	(0.114)	(0.129)	(0.125)
Percent Non-Hispanic Black	12.770***	4.986***	-7.787***
	(0.367)	(0.252)	(0.287)
Percent Hispanic	9.627***	4.163***	-5.465***
	(0.313)	(0.220)	(0.256)

Notes: The first column shows the mean absolute difference between pairs formed by matching each county in the 2010 contiguous border county sample with all possible other counties. The second column shows the mean absolute difference in the values of the variables between counties in a contiguous pair in the 2010 data. The symbol *** indicates that the mean absolute difference within county pairs is different from 0 at $p<0.01$ in columns (1) and (2). The third column represents the difference between columns (1) and (2), and *** indicates the difference between columns (1) and (2) is statistically different from 0 at $p<0.01$.